SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

CHEM 1210 – Introductory Chemistry I

1. Basic Course Information

Course Code and Title: CHE 1210 Introductory Chemistry I

Credit Hours: 4

Instructor(s): Sewite Negash and Tekalign T. Debela

Semester/Quarter:

Prerequisites:

Co-requisites:

Class Schedule:

Office Hours:

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2. Course Description

This accelerated course provides an intensive introduction to the fundamental principles of chemistry, integrating topics typically covered across two semesters into one. Major areas include chemical nomenclature, atomic structure, quantum theory, bonding theories and molecular geometries, chemical equations, oxidation-reduction, stoichiometry, thermochemistry, the states of matter, gas laws, solutions and colligative properties, periodic relationships, acid-base theories, chemical kinetics, molecular and ionic equilibria, thermodynamics, electrochemistry, and introductory nuclear and organic chemistry. Laboratory work complements lecture material and emphasizes both qualitative and quantitative analysis.

3. Course Objectives

This course is designed for students majoring in the natural sciences (chemistry, biochemistry, biology, nutrition, physics, pre-medicine), mathematics, and engineering. The objectives are to:

- Build a strong foundation in fundamental chemical principles.
- Develop laboratory and analytical skills through hands-on experiments.
- Strengthen quantitative reasoning and problem-solving abilities.
- Foster effective scientific communication, both written and oral.
- Emphasize safe laboratory practices and professional responsibility.

These objectives align with program-level educational outcomes by preparing students for success in advanced science and engineering courses, research, and pre-professional pathways.

4. Student Learning Outcomes (ABET Outcomes 1–7)

Upon successful completion of the course, students will be able to:

1. Apply foundational chemical concepts (structure of atoms, bonding, states of matter, kinetics, equilibrium, thermodynamics, electrochemistry).
2. Perform quantitative calculations and data analysis (stoichiometry, enthalpy, equilibrium constants, cell potentials; apply error analysis).
3. Conduct laboratory experiments safely and effectively (operate balances, pipettes, calorimeters, electrochemical cells; follow safety and disposal protocols).
4. Integrate theory with experimental observations (relating lab results to lecture material, interpreting data, predicting behavior).
5. Communicate scientific information clearly (prepare lab reports, use proper chemical notation, collaborate effectively in groups).
6. Apply problem-solving and critical thinking skills (design solutions, evaluate alternatives, justify conclusions using chemical principles).

5. Mapping to Program Outcomes

6. Required Textbooks and References

Chemistry: A Molecular Approach, 5th edition by Nivaldo Tro. (Pearson)

7. Course Topics and Schedule

Week	Lecture	Homework	Exam/Quizzes
1	Ch. 1–2	Ch. 1–2 problems	
2	Ch. 3	Ch. 3 problems	Quiz 1 (Ch. 1–2)
3	Ch. 4	Ch. 4 problems	
4	Ch. 5	Ch. 5 problems	Quiz 2 (Ch. 3–4)**, Exam 1 (Ch. 1–4)**
5	Ch. 6	Ch. 6 problems	
6	Ch. 7	Ch. 7 problems	Quiz 3 (Ch. 5–6)
7	Ch. 8	Ch. 8 problems	
8	Ch. 9	Ch. 9 problems	Quiz 4 (Ch. 7–8)**, Exam 2 (Ch. 5–8)**
9	Ch. 10	Ch. 10 problems	
10	Ch. 11	Ch. 11 problems	Quiz 5 (Ch. 9–10)
11	Ch. 12	Ch. 12 problems	**Exam 3 (Ch. 9–11)**
12	Ch. 13	Ch. 13 problems	Quiz 6 (Ch. 11–12) **Final Exam (Comprehensive)**

COVERED CHAPTERS

Chapter 01 | Matter, Measurement, And Problem Solving
Chapter 02 | Atoms And Elements
Chapter 03 | Molecules And Compounds
Chapter 04 | Chemical Reactions & Chemical Quantities
Chapter 05 | Introduction To Solutions & Aquesous Reactions
Chapter 06 | Gases
Chapter 07 | Thermochemistry
Chapter 08 | The Quantum-Mechanical Model Of The Atom
Chapter 09 | Periodic Properties Of The Elements
Chapter 10 | Chemical Bonding I: The Lewis Model
Chapter 11 | Chemical Bonding II: Molecular Shapes, Valence Bond Theory, And Molecular Orbital Theory
Chapter 12 | Liquids, Solids, And Intermolecular Forces
Chapter 13 | Solids And Modern Materials

8. Laboratory Component

Students will engage in a variety of hands-on laboratory activities designed to reinforce lecture concepts. They will perform experiments such as measuring density and precision, analyzing chemical reactions, titrations, investigating gas laws, thermochemistry, kinetics, equilibrium, acids and bases, thermodynamics, and electrochemistry. Equipment includes balances, pipettes, burettes, calorimeters, hot plates, thermometers, and molecular model kits. Safety is emphasized with required PPE (lab coat, goggles, gloves) and proper handling and disposal of chemicals. Expected outcomes include proficiency with laboratory techniques, accurate data collection, critical analysis, and integration of theory with experimental evidence.

9. Grading Policy

Grades are based on performance in lecture and lab, both of which must be passed to receive credit.

Breakdown:

- Homework (12): 15%
- Quizzes (6, drop 1) : 25 %
- Exam/Final (4, drop 1): 60 %

Scale:

- A = 90–100%
- B = 80–89.9%
- C = 70–79.9%
- D = 60–69.9%
- F = < 60%

10. Academic Integrity

State the academic honesty policy and consequences in accordance with institutional regulations.

Academic honesty is expected at all times. Cheating, plagiarism, fabrication of data, or unauthorized collaboration will result in an **F** on the assignment and may result in an **F** for the course, with possible reporting to the administration. Exams prohibit cell phones, messaging devices, and programmable calculators.

The use of artificial intelligence (AI) tools (such as ChatGPT, Copilot, etc.) must follow the instructor's explicit guidelines. Using AI to generate lab reports, problem solutions, or written assignments without proper acknowledgment is considered plagiarism and a violation of academic integrity. Students may use AI for study support or brainstorming only if it is clearly permitted, and they remain fully responsible for the accuracy and originality of their submitted work.

11. Accessibility and Student Support

Information on disability services, tutoring, counseling, and academic support resources.

Students requiring accommodations should contact the institution's Special Programs and Services (SPS ???). Additional support resources include tutoring, counseling, and academic advising services, available through the academic success center.

12. Safety Considerations

Outline safety guidelines for labs, fieldwork, or other activities.

Students must wear PPE at all times in the lab, handle chemicals with care, and follow waste disposal and emergency protocols. They must know the locations of safety showers, eyewash stations, fire extinguishers, and exits. By following these practices, students will develop professional laboratory habits while minimizing risk.